Summarizing Data

Module 1: Python Foundations

```
games = [("Clue", 1)]
for title, n in games:
    if n < 3:
        print("Low:", title)
```

By the End of Today, You'll Be Able To…

1

Write a second function with a lighter template

2

Keep validating and summarizing separate

You'll Know This by the End of Today

SUMMARIZING

Turning already-valid data into a useful report.

Reading Recap: One Job at a Time

Today's Reading: A Second, Single-Job Function

A fitness tracker already has `validate_days`, which keeps only the good days. Now you want a weekly summary. Should that same function do it, or should the summary get its own? Reopen the reading:

One Job at a Time: Writing a Second, Single-Job Function

The Scenario: A Week of Step Counts

A week of steps

Mon 6,200. Tue 9,120. Wed 8,000. Thu 4,300. Fri 10,450. `validate_days` already kept only good days.

The new request

A weekly summary with total steps and how many days met the 8,000-step goal.

One new function

`summarize_steps` returns `{'total': 38070, 'goal_days': 3}`. It never re-checks a day.

READING RECAP

Key Terms and Code From the Reading

Accumulator

`total = 0` before the loop. `total += steps` adds each day. After the loop, total is 38,070.

One job

`validate_days` checks. `summarize_steps` summarizes. Each job fits in one sentence.

Return value

`return` hands the summary dict back. Printing or saving it is a separate job.

READING CHECK

The week: 6,200, 9,120, 8,000, 4,300 and 10,450 steps. What is `goal_days` for `summarize_steps(days, 9000)`?

A 3

B 2

C 1

ANSWER KEY

Answer Key: Change the Goal, Not the Function

Correct answer: B. 2

Only Tue (9,120) and Fri (10,450) reach 9,000. Wed's 8,000 met the 8,000 goal, but not this one. Only the argument changed, not the function.

READING CHECK

The app now needs to email each user their weekly summary. What should the team do?

A Write a separate email function

B Add the email code to the end of the summary function

C Re-check every step count before emailing

ANSWER KEY

Answer Key: A New Job Gets a New Function

Correct answer: A. A separate email function

`summarize_steps` already returns the summary, so a new function can take that dict and send it. `summarize_steps` stays the same, and nothing re-checks the data.

Your Questions From the Reading

Pair up

1 min: tell a partner one question you still have.

Ask

We take 2 or 3 questions now.

Park it

Bigger ones wait for today's matching slide.

A Different Job Entirely

`validate_batch` ends when it says valid or not. Today's function starts from there.

IN THE NEWS

A Famous Average Silently Left Out 5 Countries



World Affairs Council of Philadelphia talk on the error, October 17, 2013. Photo: World Affairs Council of Philadelphia, CC BY 2.0, via Wikimedia Commons

The authors' formula skipped rows 45-49
UMass researchers caught it while replicating
Authors admitted the error, kept their claim

A summary only reports on the rows it is handed. How would your code make sure it covers every row?

Source: Herndon, Ash & Pollin, PERI Working Paper 322, April 2013

Validating and Summarizing Are Different Jobs

VALIDATE

Decide valid or not.

Validating and Summarizing Are Different Jobs

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Decide valid or not.

THEN

Pass approved data forward.

Validating and Summarizing Are Different Jobs

VALIDATE

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SUMMARIZE

Report what's in the valid data.

Two Librarians, Two Jobs

RETURNS DESK

Check whether a returned book is damaged.
That's the whole job.

NEW ARRIVALS

Use only accepted books to update the shelf.
A separate job.

CHECK FOR UNDERSTANDING

What stands in for `summarize_batch` in the library analogy?

A Checking a returned book

B Both jobs by one person

C Updating the new-arrivals shelf

ANSWER KEY

Library Analogy Recall

Correct: Updating the new-arrivals shelf

That job uses only books already accepted — just like `summarize_batch` works only with data `validate_batch` approved.

Two Functions, Two Jobs

`validate_batch`

Decides valid or not, per line.
Reports exactly what's wrong.

`summarize_batch`

Assumes data is already valid.
Reports what's there, formatted.

IN COMMON: each function owns one clear job.

CODE ALONG

Open the Code-Along

Open `Module1_Lesson11_summarizing-data-with-a-second-function_STUDENT.py`
Run it after every section: `python3` + the file name

Sections, in order:

1. Step 1: A Formatted Listing
   Start with the simpler half: a formatted line per record.
2. Step 2: Add a Low-Availability Alert
   Now add the second half: flagging anything running low.
3. Step 3: Add a Total Count
   Add a quick total: the kind of number a stakeholder often asks for first.

Step 1: A Formatted Listing

```python
games = [
    ("Chess", 4),
    ("Checkers", 6),
    ("Clue", 1),
    ("Uno", 8),
]
def summarize_batch(games):
    for title, players in games:
        print(f"{title}: {players} available")

summarize_batch(games)
```

Step 2: Add a Low-Availability Alert

python

```python
def summarize_batch(games):
    for title, players in games:
        print(f"{title}: {players} available")
    print()
    print("Low availability:")
    for title, players in games:
        if players < 3:
            print(f"  {title} ({players})")

summarize_batch(games)
```

Step 3: Add a Total Count

python

```python
def summarize_batch(games):
    for title, players in games:
        print(f"{title}: {players} available")
    print()
    print("Low availability:")
    for title, players in games:
        if players < 3:
            print(f"  {title} ({players})")
    print()
    print(f"{len(games)} games total")
summarize_batch(games)
```

CODE-ALONG DONE

Back to the Slides: Review

You just typed it and ran it. Now check that it stuck, without the code in front of you.

QUICK CHALLENGE

What’s the third line the Step 1 code prints?

A

```
"Clue: 1 available"
```

B

```
"Checkers: 6 available"
```

C

```
"Uno: 8 available"
```

ANSWER KEY

Formatted Listing Trace

Correct: `"Clue: 1 available"`

The loop prints in list order: Chess, Checkers, Clue third, then Uno.

Watch It Run: Every Row, Every Time

Step 1 of 6

We call summarize_batch with all 4 games

```
games = [("Chess", 4), ("Checkers", 6),
         ("Clue", 1), ("Uno", 8)]
def summarize_batch(games):
    for title, players in games:
        print(f"{title}: {players} available")
summarize_batch(games)
```

OUTPUT

ANIMATED IN CLASS: watch it run in the lecture recording

QUICK CHALLENGE

In the Step 2 code, what changes if Uno had 2 players?

A Nothing changes

B Uno also appears in low availability

C The function crashes

ANSWER KEY

Alert List Trace

Correct: Uno also appears in low availability

`2 < 3` is `True`, so Uno prints in the alert section too. Every matching game is reported.

QUICK CHALLENGE

Add one more game to the Step 3 code. What prints last?

A

`"5 games total"`

B

`"4 games total"`

C

It stays the same

ANSWER KEY

Total Count Trace

Correct: `"5 games total"`

`len(games)` reflects the list's current size automatically. Add one tuple and the total becomes 5.

TIP

“One function or two? Defend your choice.”

COMBINE

One function validates and summarizes.

SEPARATE

`validate_batch` + `summarize_batch` each keep one job.

Answer with QAR: your first question, your assumption, the risk.

MODEL ANSWER

Model Answer: QAR

QUESTION

Is this a one-off script, or code we'll rerun on every batch?

ASSUMPTION

We'll rerun it on every new batch file, and keep changing it.

RISK

Combined, a summary tweak can quietly break a validation rule.

Stakeholders Want the Summary, Not Raw Rows

COMMON TASK

Turning raw data into a report happens constantly.

USEFUL FIRST

Stakeholders want the summary.

SHAREABLE

A single-job function is easy to reuse.

QUICK CHALLENGE

`summarize_batch` adds a `players < 0` check. What's wrong?

A Nothing, it's extra safety

B It crashes on an int

C It repeats validation's job

ANSWER KEY

Don’t Re-Validate in `summarize_batch`

Correct: It’s redoing `validate_batch`’s job

`validate_batch` already rejected negatives: `"-1".isdigit()` is `False`. Re-checking here blurs the two jobs together.

TURN AND TALK

What Else Might a Stakeholder Want?

COUNT

How many records?

ALERT

Which items need attention?

YOUR IDEA

What other summary would be useful?

Key Terms — Recap

SUMMARIZING

Reporting useful information from data that has already been validated.

So Far, We've Covered…

TODAY

A second function, a different job.

PROJECT

This is the project's own `print_summary`.

NEXT

Practice all of it together.